

C++ Programming Exam - Answer Sheet

Student: Alex Chen | ID: 2024005

Question 1 (15 points)

- 1.1 Which line of the above code is both a definition and an initialization?
Statement c (double d = 9;)
- 1.2 Which line of the above code declares a variable without allocating memory for it?
Statement a (extern int x;)
- 1.3 Which lines of the above code are assignments?
Statements d and e (a = 1; and b = 5;)
- 1.4 According to the following statement: const int *p;
the value of the integer that p points to cannot change
- 1.5 When is the following logical test true? (x>=y && ! x && x* y < 0 && y==0)
never
- 1.6 Which of the following cases of mixed expressions is/are correct?
'f' - 'd' is an int
- 1.7 Given: char c='b'; char *pc=&c; char *&rc=pc; then after (*rc)++; what happens?
it increases *rc
- 1.8 Which of the provided statements (if any) are invalid?
px =&x; double ■ = *px ;, px = px = rx;, px = py = *x;
- 1.9 Which of the provided statements would give the value of x?
**ppx, *(static_cast (pp)), *(&px;)), *((double*)pp)
- 1.10 Which expressions give as result an int equal to 6?
'z' - 't', 13 % 7, 55 % 7
- 1.11 What will be the value of x after: int x = (7>6 ? 1+8 : 8)
8
- 1.12 Which of the following(s) is a (are) valid function declaration?
void func(int x);, void func(int);
- 1.13 Which function will be called by: printFun(2.0*f) given float f;
void printFun(double)
- 1.14 How many times is function fib called when num is 3?

5

1.15 In a function with return type void, what happens at return?

No value is returned

Question 2 (5 points)

2.1 $77 + c + i + 1L$

long

2.2 $6.55f + f / 1.5 - 9 / 8$

double

2.3 $'z' - 'z'$

int

2.4 $b + c$

int

2.5 $'t' - 'a' + c$

int

2.6 $77.8f + 4 * 0.5f + 45L$

float

2.7 $42L + (int) d + 94.3f + int(4.9)$

float

2.8 $0.0 + f + c$

double

2.9 $5.28L * d * 3 + 4.5$

long double

2.10 $1.5f / d * f + 6.9 * 4L$

double

Question 3 (10 points)

Replace the comments with expressions:

```
// Local name: cout << "\n Local name : " << name; // Global name: cout << "\n Global  
name : " << ::name; // Object member name: cout << "\n Object name : " << m.name; //  
Invoke member function print(): m.print();
```

Question 4 (10 points)

4.1 Fragment 1 output

x is 10 *y is 10 **z is 10

4.2 Fragment 2 output

a is 1 b is 1 c is 1 x is 1 y is 0 z is 1

4.3 Fragment 3 output

x[0] = 1 y[0] = 2

4.4 Fragment 4 output

test1=20test2=50.8test3=68

4.5 Fragment 5 output

At the beginning a = 10 b = 20 Before func2 a = 5 b = 40 After func2 a = 40 b = 5 At the end a = 20 b = 10

Question 5 (40 points)

Write a program that prints an hourglass figure:

```
#include <iostream> using namespace std; int main() { int n; if (!(cin >> n)) { cerr << "Error: invalid input." << endl; return EXIT_FAILURE; } if (n <= 1 || (n % 2 == 0)) { cout << "Error: n must be an odd integer greater than 1." << endl; return EXIT_FAILURE; } int mid = n / 2; for (int r = 0; r <= mid; ++r) { int stars = n - 2*r; if (stars == n) { for (int k = 0; k < n; ++k) cout << '*'; } else if (stars == 1) { cout << '*'; } else { cout << '*'; for (int k = 0; k < stars - 2; ++k) cout << ' '; cout << '*'; } cout << "\n"; } for (int r = mid - 1; r >= 0; --r) { int stars = n - 2*r; if (stars == n) { for (int k = 0; k < n; ++k) cout << '*'; } else if (stars == 1) { cout << '*'; } else { cout << '*'; for (int k = 0; k < stars - 2; ++k) cout << ' '; cout << '*'; } cout << "\n"; } return EXIT_SUCCESS; }
```

Question 6 (20 points)

Provide the missing constructor and methods:

```
Complex(double real = 0.0, double imaginary = 0.0) : real(real), imaginary(imaginary) {}  
double Complex::get_real(void) { return real; }  
double Complex::get_imaginary(void) { return imaginary; }  
void Complex::set_real(double r) { real = r; }  
void Complex::set_imaginary(double i) { imaginary = i; }
```